

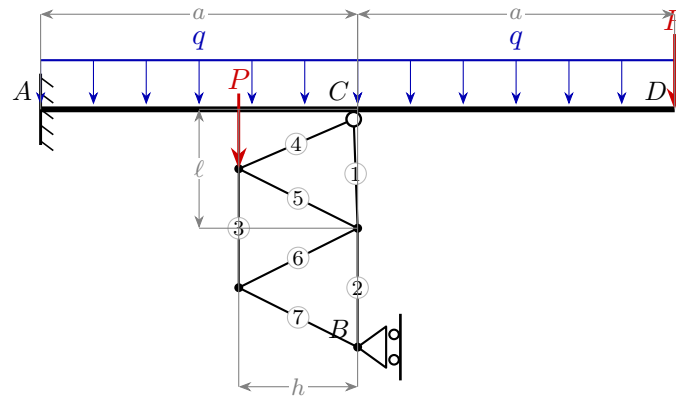
University of Natural Resources and Life Sciences, Vienna (BOKU)

VU Mechanik – LAWI 100515 (PI)

Exam Group A

Date: 26.06.2026

Model solution



1. Static determinacy

Counting criterion $n = 3k - (a + z)$ (bodies k , reactions a , interaction forces z):

$$n = 3k - (a + z) = 3 \cdot 2 - (4 + 2) = 0 \Rightarrow \text{statically determinate}$$

2. Support reactions and hinge forces

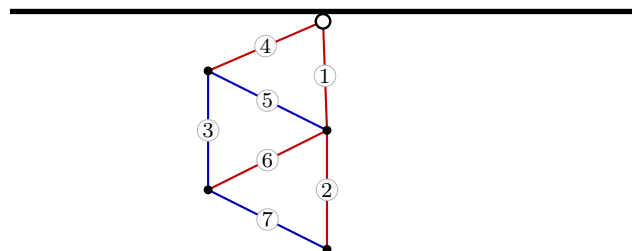
$$A: R_x = 5, R_y = 52, M = 248 \quad B: R_x = -5, R_y = 0$$

$$C: H_x = -5, H_y = -10$$

3. Truss member forces

Member	Force S [kN]	Type
1	7.5	tension
2	2.5	tension
3	-5	compression
4	5.59	tension
5	-5.59	compression
6	5.59	tension
7	-5.59	compression

red: tension (+) blue: compression (-)



4. Section-force diagrams of the beams

